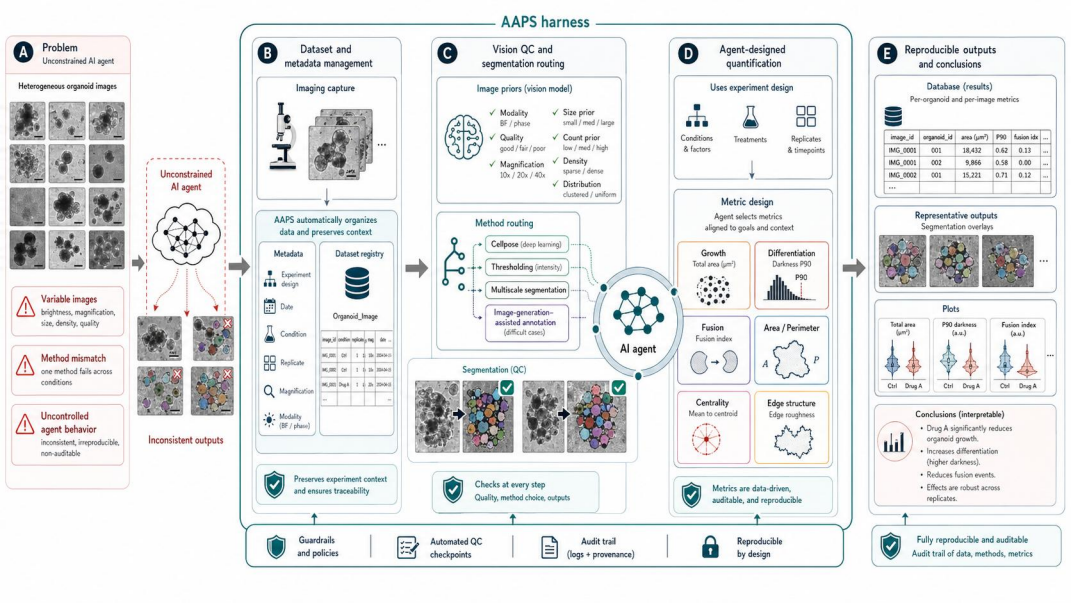

AAPS harness
A
Problem
Unconstrained AI agent
Heterogeneous organoid images
Variable images
brightness, magnification, size, density, quality
Method mismatch
one method fails across conditions
Uncontrolled agent behavior
inconsistent, irreproducible, non-auditable
Unconstrained AI agent
Inconsistent outputs
B
Dataset and metadata management
Imaging capture
AAPS automatically organizes data and preserves context
Metadata
Experiment design
Date
Condition
Replicate
Magnification
Modality (BF / phase)
Dataset registry
Organoid_Image
Preserves experiment context and ensures traceability
C
Vision QC and segmentation routing
Image priors (vision model)
Modality BF / phase
Quality good / fair / poor
Magnification 10x / 20x / 40x
Size prior small / med / large
Count prior low / med / high
Density sparse / dense
Distribution clustered / uniform
Method routing
Cellpose (deep learning)
Thresholding (intensity)
Multiscale segmentation
Image-generation–assisted annotation (difficult cases)
Segmentation (QC)
Checks at every step
Quality, method choice, outputs
AI agent
D
Agent-designed quantification
Uses experiment design
Conditions & factors
Treatments
Replicates & timepoints
Metric design
Agent selects metrics aligned to goals and context
Growth
Total area (µm²)
Differentiation
Darkness P90
P90
Fusion
Fusion index
Area / Perimeter
A
P
Centrality
Mean to centroid
Edge structure
Edge roughness
Metrics are data-driven, auditable, and reproducible
E
Reproducible outputs and conclusions
Database (results)
Per-organoid and per-image metrics
image_id
organoid_id
area (µm²)
P90
fusion idx
IMG_0001
001
18,432
0.62
0.13
IMG_0001
002
9,866
0.58
0.00
IMG_0002
001
15,221
0.71
0.12
Representative outputs
Segmentation overlays
Plots
Total area (µm²)
P90 darkness (a.u.)
Fusion index (a.u.)
Ctrl
Drug A
Conclusions (interpretable)
Drug A significantly reduces organoid growth.
Increases differentiation (higher darkness).
Reduces fusion events.
Effects are robust across replicates.
Fully reproducible and auditable
Audit trail of data, methods, metrics
Guardrails and policies
Automated QC checkpoints
Audit trail (logs + provenance)
Reproducible by design